

Test-Time Adaptation: What Is Allowed to Change?

At test time, a system can change different parts of its computation - such as model parameters, normalization behavior, prompts, inputs, or temporary state - and those choices create different adaptation costs, stability risks, and memory behavior; adaptation must therefore be distinguished from simply spending more inference compute.

Why this problem exists

A deployed model may encounter data, tasks, or operating conditions that differ from training. Test-time adaptation (TTA) uses information available at inference to modify some component of the prediction process. The important distinction is the adaptation surface. EATA updates selected model parameters while trying to reduce forgetting [R1]; TAST trains lightweight adaptation modules using nearest-neighbor-informed pseudo-labels [R2]; Test-Time Prompt Tuning adapts prompts from a test sample while preserving the pretrained vision-language model [R3]. A 2024 survey groups TTA methods by changes to the model, inference, normalization, sample, or prompt [R6].

Mechanism and trade-offs

Parameter adaptation can be flexible but requires backward computation and can drift or forget. EATA explicitly targets prediction cost and forgetting [R1]. Online adaptation can also collapse under noisy or imbalanced streams; SAR studies these instabilities [R4]. DeYO shows that entropy alone can be an unreliable confidence signal in biased settings [R5]. Less invasive adaptation surfaces - prompts, normalization statistics, input transformations, or temporary modules/state - can reduce the amount of model machinery that changes, but they also constrain what can be adapted.

Do not confuse adaptation with test-time compute

The artifact separates adaptation from inference strategy. A system may spend more inference compute without adapting its weights or memory: self-consistency samples multiple reasoning paths and aggregates answers [R10]; Tree of Thoughts searches over candidate thoughts with evaluation and backtracking [R9]; repeated sampling is itself an inference-scaling axis [R11]. Snell et al. show that useful test-time-compute allocation depends on problem difficulty [R12]. Latent approaches move the extra compute away from visible reasoning tokens: Coconut feeds continuous hidden reasoning states back into the model [R13], while recurrent-depth models unroll a recurrent block further at test time [R14].

BDH / BDH-CQ connection

BDH-CQ is used as a case study rather than presented as a classical distribution-shift TTA method. Its authors report that inference-time inputs continuously update recurrent memory and that queries are solved through iterative latent computation [R8]. The earlier BDH paper describes a brain-inspired architecture with locally interacting units and inference-time synaptic working-memory concepts [R7]. The artifact must label these as

published claims and must not imply that its teaching animations are literal BDH state traces or independent reproductions.

Strengths, limits, and evidence maturity

TTA is valuable because deployment is rarely identical to training, but adaptation introduces a new failure surface: the system can change itself in response to unreliable test evidence. Current work therefore focuses not only on whether to adapt, but on sample selection, regularization, confidence, stability, and which component to expose for change [R1-R5]. BDH/BDH-CQ provide a recent architecture-native state-based case study, but their reported results should be separated from independently reproduced evidence. The artifact is strongest when the learner can manipulate a real variable, see the consequence, compare output to an expected result, and intentionally reach a failure case.

Representative primary sources

[R1] Niu et al. (2022) - Efficient Test-Time Model Adaptation without Forgetting (EATA). <https://arxiv.org/abs/2204.02610>

[R2] Jang, Chung & Chung (2022) - Test-Time Adaptation via Self-Training with Nearest Neighbor Information (TAST). <https://arxiv.org/abs/2207.10792>

[R3] Shu et al. (2022) - Test-Time Prompt Tuning for Zero-Shot Generalization in Vision-Language Models (TPT). <https://arxiv.org/abs/2209.07511>

[R4] Niu et al. (2023) - Towards Stable Test-Time Adaptation in Dynamic Wild World (SAR). <https://arxiv.org/abs/2302.12400>

[R5] Lee et al. (2024) - Entropy is not Enough for Test-Time Adaptation: From the Perspective of Disentangled Factors (DeYO). <https://arxiv.org/abs/2403.07366>

[R8] Engdahl et al. (2026) - BDH-CQ: In-Context Learning with Recurrent Latent Reasoning. <https://arxiv.org/abs/2608.09888>